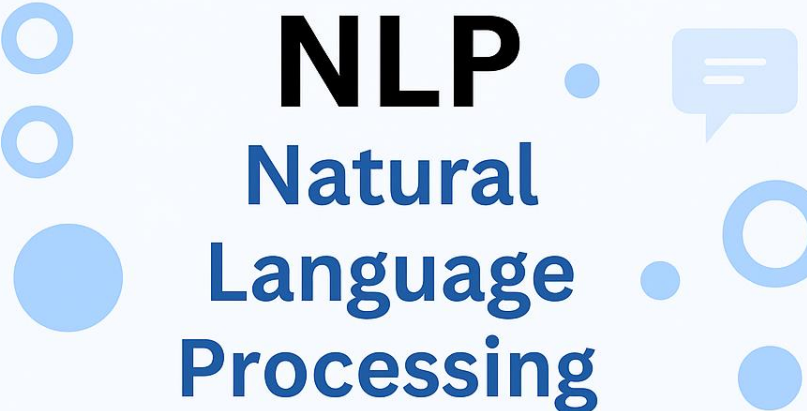
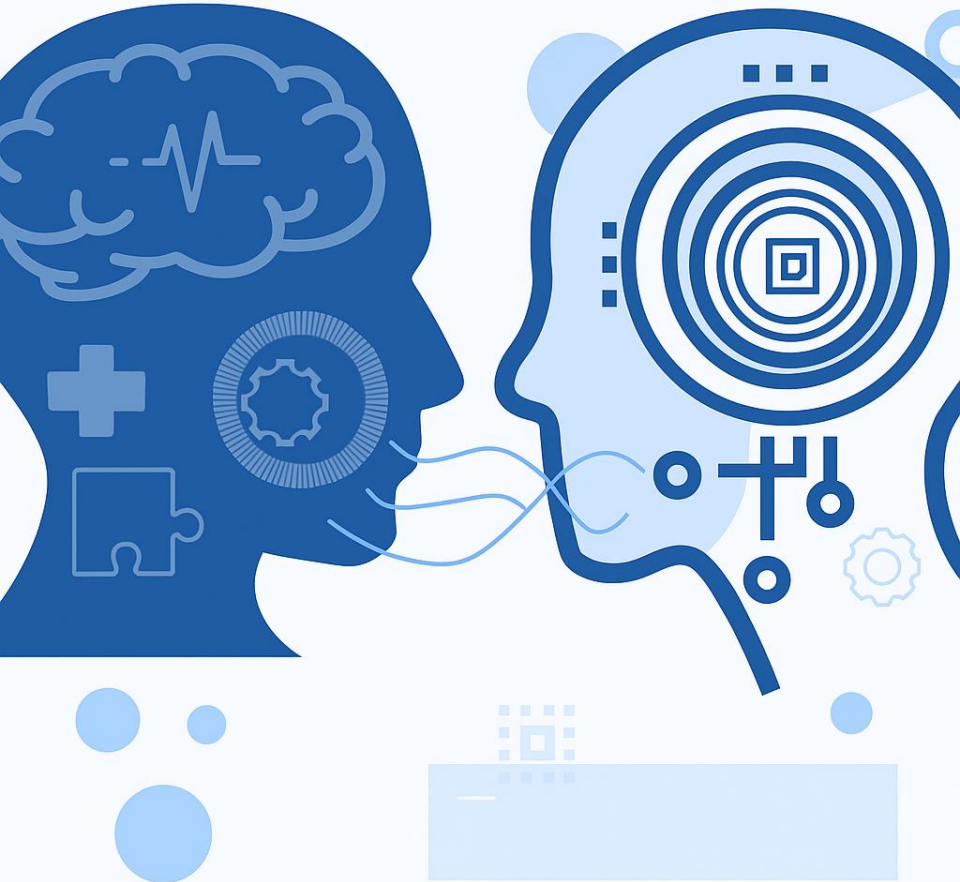




NLP

Natural
Language
Processing



NATURAL LANGUAGE PROCESSING (NLP)

PMDS606L

MODULE 4

LECTURE 4

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VITERBI ALGORITHM EXAMPLE

Hidden States (POS Tags):

DET = Determiner

NOUN = Noun

VERB = Verb

Observations (Words):

The, dog, runs

Transition Probabilities (between tags)

$$P(DET \rightarrow NOUN) = 0.9, \quad P(DET \rightarrow VERB) = 0.1$$

$$P(NOUN \rightarrow VERB) = 0.8, \quad P(NOUN \rightarrow NOUN) = 0.1$$

$$P(VERB \rightarrow NOUN) = 0.2 \quad (\text{small probability})$$

Emission Probabilities (word | tag)

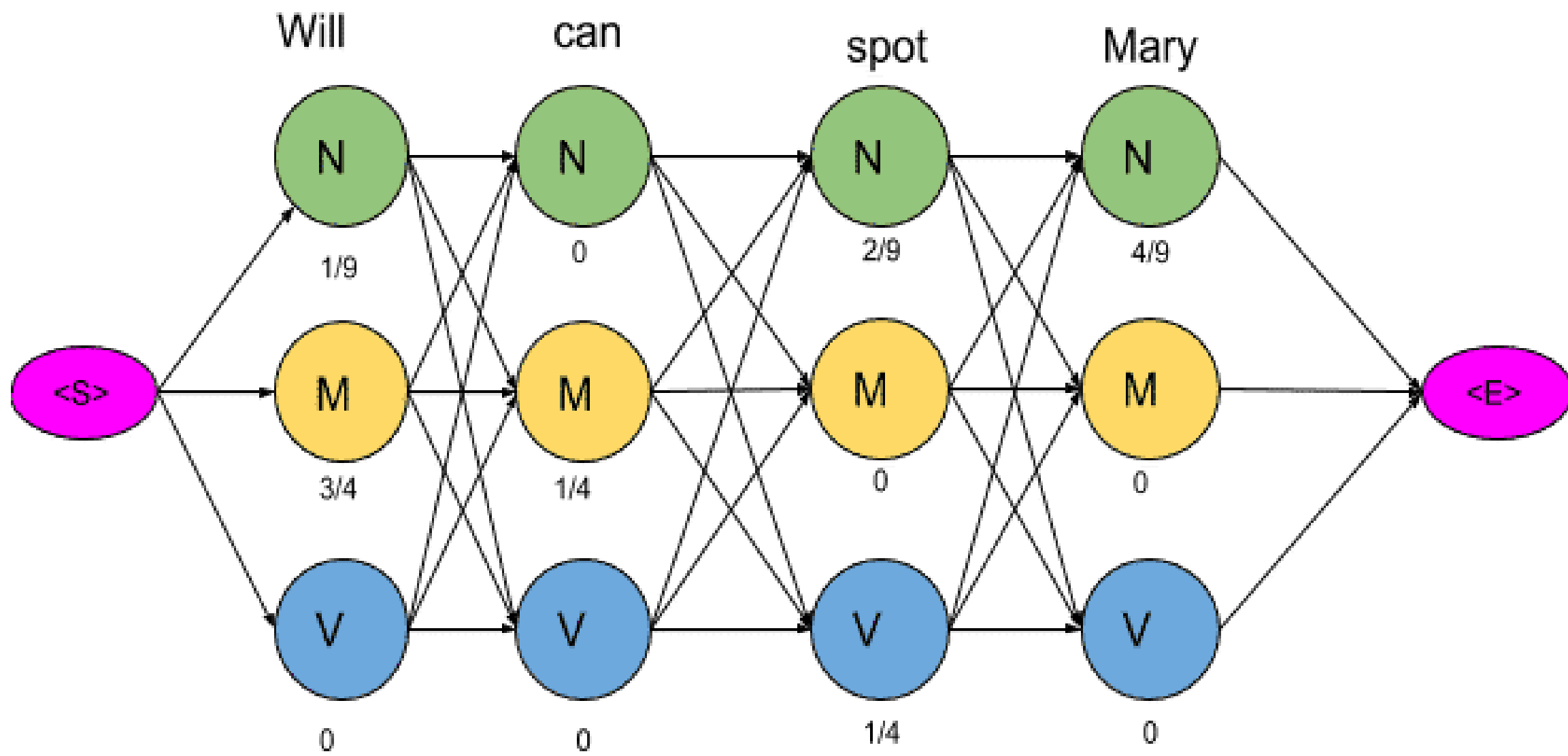
$$P(\text{"The"} \mid DET) = 1.0$$

$$P(\text{"dog"} \mid NOUN) = 0.9$$

$$P(\text{"runs"} \mid VERB) = 0.95$$

Words	Noun	Model	Verb
Mary	4/9	0	0
Jane	2/9	0	0
Will	1/9	3/4	0
Spot	2/9	0	1/4
Can	0	1/4	0
See	0	0	2/4
pat	0	0	1

	N	M	V	<E>
<S>	3/4	1/4	0	0
N	1/9	3/9	1/9	4/9
M	1/4	0	3/4	0
V	4/4	0	0	0



	NNP	MD	VB	JJ	NN	RB	DT
$\langle s \rangle$	0.2767	0.0006	0.0031	0.0453	0.0449	0.0510	0.2026
NNP	0.3777	0.0110	0.0009	0.0084	0.0584	0.0090	0.0025
MD	0.0008	0.0002	0.7968	0.0005	0.0008	0.1698	0.0041
VB	0.0322	0.0005	0.0050	0.0837	0.0615	0.0514	0.2231
JJ	0.0366	0.0004	0.0001	0.0733	0.4509	0.0036	0.0036
NN	0.0096	0.0176	0.0014	0.0086	0.1216	0.0177	0.0068
RB	0.0068	0.0102	0.1011	0.1012	0.0120	0.0728	0.0479
DT	0.1147	0.0021	0.0002	0.2157	0.4744	0.0102	0.0017

	Janet	will	back	the	bill
NNP	0.000032	0	0	0.000048	0
MD	0	0.308431	0	0	0
VB	0	0.000028	0.000672	0	0.000028
JJ	0	0	0.000340	0	0
NN	0	0.000200	0.000223	0	0.002337
RB	0	0	0.010446	0	0
DT	0	0	0	0.506099	0

